

Lattice — Feature Showcase

Open this file with Lattice to see every feature rendered live.
Switch between **Edit**, **Preview**, and **Dual** view with the toolbar selector.
Try `Ctrl/Cmd+Shift+T` to auto-generate the Table of Contents below.

Table of Contents

- [Table of Contents](#)
- [Text Formatting](#)
- [Highlight Marks](#)
- [Lists](#)
 - [Unordered](#)
 - [Ordered](#)
 - [Definition-style \(using bold + indent\)](#)
- [Task Lists \(GFM\)](#)
- [Tables](#)
- [Code](#)
 - [Inline code](#)
 - [Fenced code blocks](#)
- [Blockquotes](#)
 - [GitHub-style Alerts \(rendered by GitHub and Lattice\)](#)
- [Mathematics — KaTeX](#)
 - [Inline math](#)
 - [Display math — Calculus](#)
 - [Display math — Linear Algebra](#)
 - [Display math — Series & Limits](#)
- [Mermaid Diagrams](#)
 - [Flowchart — Lattice Edit Cycle](#)
 - [Sequence Diagram — File Open Flow](#)
 - [Class Diagram — Core Data Model](#)
 - [Pie Chart — Supported Platforms](#)
- [Images](#)
- [Links](#)
- [HTML in Markdown](#)
- [Keyboard Shortcuts Reference](#)

Text Formatting

Normal paragraph text flows here. Lattice renders Markdown as **close to GitHub** as possible, so files look identical whether viewed in the editor preview, on GitHub, or printed.

Style	Syntax	Result
Bold	<code>**bold**</code>	bold
Italic	<code>*italic*</code>	<i>italic</i>
Bold + italic	<code>***both***</code>	<i>both</i>
Strikethrough	<code>~~strikethrough~~</code>	strikethrough
Inline code	<code>`code`</code>	<code>code</code>
Superscript	<code>X<sup>2</sup></code>	X ²
Subscript	<code>H<sub>2</sub>O</code>	H ₂ O

Note — sub/superscript: CommonMark/GFM has no native sub/superscript syntax. Lattice renders `<sup>` and `<sub>` HTML tags directly (GitHub and Obsidian do too). We recommend, though that you use alternatives that work everywhere: KaTeX math ($x^2 \rightarrow X^2$, $H_{20} \rightarrow H_2O$) or Unicode characters (x^2 , H_2O). The linter flags these tags as a portability hint — they may not render in all renderers.

Highlight Marks

Highlight marks are a Lattice-exclusive feature. Wrap any text in `==double equals==` to render it with a **bright amber background**.

This works in **any context** — inside a sentence, `==across italic and bold spans==`, or on its own line:

This entire sentence is highlighted.

[!NOTE] Highlight marks are **not** rendered inside code blocks or inline code — `==this stays raw==` .

Lists

Unordered

- First item
- Second item
 - Nested item A
 - Nested item B
 - Deeply nested
- Third item with **bold** and *italic* text

Ordered

1. Install Lattice from the [releases page](#)
2. Open any `.md` file — double-click or via **File** → **Open**
3. Choose your view:
 - i. **Edit** — raw CodeMirror editor only
 - ii. **Preview** — rendered HTML only
 - iii. **Dual** — side-by-side (four layout variants)
4. Print with `Ctrl/Cmd+P` — always renders to a clean light theme

Definition-style (using bold + indent)

Local-first All files stay on your machine. No cloud sync required.

Documents as Code Plain-text `.md` files are diff-friendly, version-controlled, and AI-ready.

Task Lists (GFM)

GitHub Flavored Markdown task lists render as interactive checkboxes in Lattice.

- ☒ **Install Lattice**
- ☒ **Open the demo file**
- ☒ **Explore Edit / Preview / Dual views**
- ☐ Try `Ctrl/Cmd+Shift+T` to regenerate the Table of Contents
- ☐ Try `Ctrl/Cmd+Shift+L` to auto-align a pipe table
- ☐ Paste a screenshot with `Ctrl/Cmd+V` — it auto-saves to a sibling folder
- ☒ **Open a second file in a new window (multi-window support)**
- ☐ Try printing to PDF with `Ctrl/Cmd+P`

Tables

Lattice supports **GFM pipe tables** with left, center, and right alignment. Press `ctrl/cmd+shift+L` with the cursor anywhere in the table to auto-format it.

Feature	Edit View	Preview View	Dual View
CodeMirror editor	✓	—	✓
Live rendered preview	—	✓	✓
Scroll sync	—	—	✓
Print (Ctrl+P)	✓	✓	✓
Highlight marks	✓	✓	✓
Mermaid diagrams	—	✓	✓
KaTeX mathematics	—	✓	✓

Code

Inline code

Use backticks for `inline code`. Great for `file/paths`, `commands`, and `variable_names`.

Fenced code blocks

Lattice passes code fences to the preview unchanged — syntax highlighting is provided by the renderer.

```
// Lattice highlight-mark plugin – converts ==text== to <mark> elements
export const MARK_RE = /==(?:[^\n]|(?:!|=))+?)/g;

export function splitAtMarks(text: string): HastNode[] {
  const parts: HastNode[] = [];
  let lastIndex = 0;
  MARK_RE.lastIndex = 0;
  let match: RegExpExecArray | null;
  while ((match = MARK_RE.exec(text)) !== null) {
    if (match.index > lastIndex)
      parts.push({ type: 'text', value: text.slice(lastIndex, match.index) });
    parts.push({
      type: 'element', tagName: 'mark',
      properties: {},
      children: [{ type: 'text', value: match[1] }],
    });
    lastIndex = match.index + match[0].length;
  }
  if (lastIndex < text.length)
    parts.push({ type: 'text', value: text.slice(lastIndex) });
  return parts;
}

// Tauri command – read file with conflict detection
#[tauri::command]
pub async fn read_file(path: String) -> Result<FileReadResult, String> {
  let content = fs::read_to_string(&path)
    .map_err(|e| format!("Cannot read {path}: {e}"))?;
  let hash = compute_sha256(&content);
  Ok(FileReadResult { content, hash })
}

# Build a production release
export LATTICEBUILD_NO="MyBuild-001"
npm run tauri build
```

Blockquotes

Single-level blockquote. Great for callouts, citations, or side notes.

Nested blockquotes are also supported:

Inner quote — *"Documents as code, not documents as data."*

— Lattice design philosophy

GitHub-style Alerts (rendered by GitHub and Lattice)

[!NOTE] Useful information that users should know, even when skimming.

[!TIP] Optional advice for a better experience.

[!IMPORTANT] Key information users need to know to achieve their goal.

[!WARNING] Urgent information that needs immediate attention.

[!CAUTION] Advises about risks or negative outcomes.

Mathematics — KaTeX

Lattice renders mathematics via **KaTeX**. Use `$...$` for inline and `$$...$$` for display math.

Inline math

Einstein's famous equation $E = mc^2$ and the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ render inline.

The Euler identity $e^{i\pi} + 1 = 0$ is often called the most beautiful equation in mathematics.

Display math — Calculus

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$
$$\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

Display math — Linear Algebra

$$\mathbf{Ax} = \lambda \mathbf{x} \implies \det(\mathbf{A} - \lambda \mathbf{I}) = 0$$
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

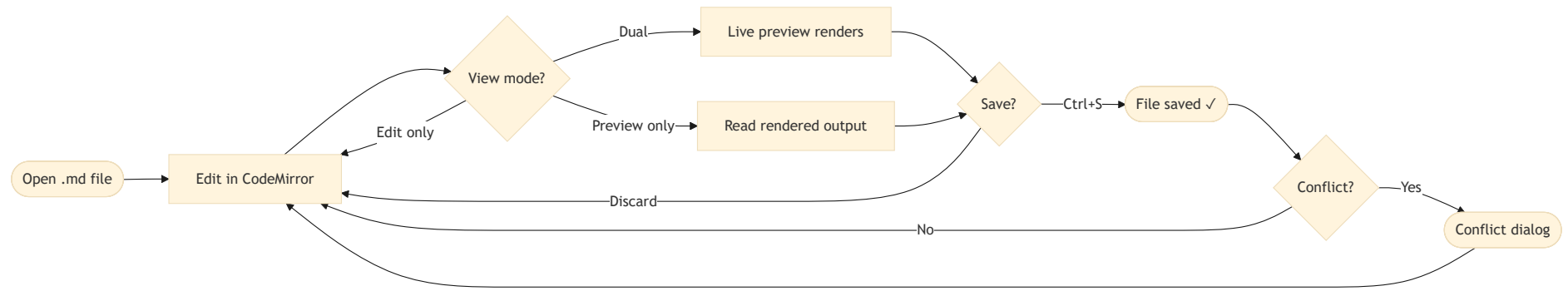
Display math — Series & Limits

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$
$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

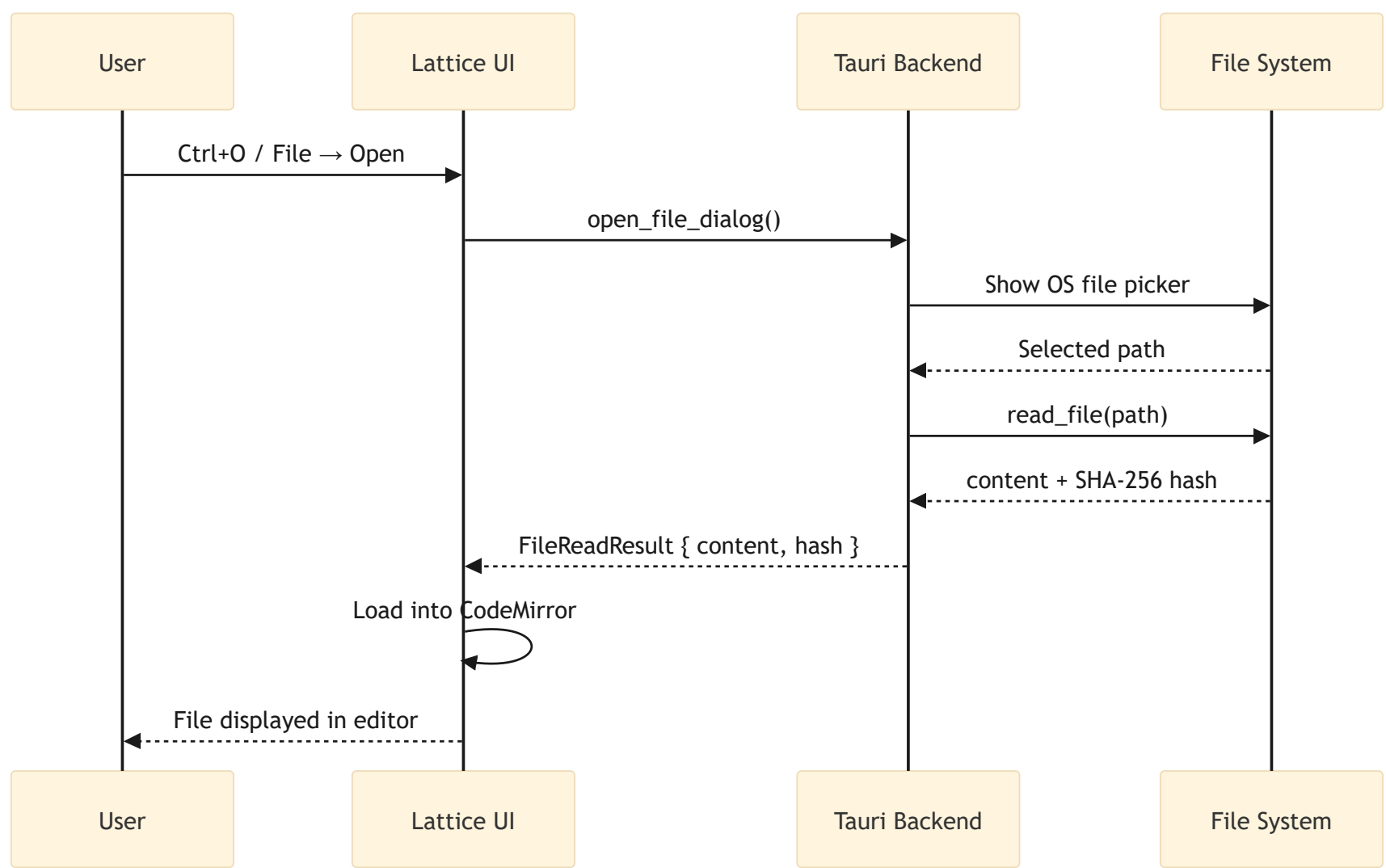
Mermaid Diagrams

Lattice renders ````mermaid```` fenced blocks inline in the preview.

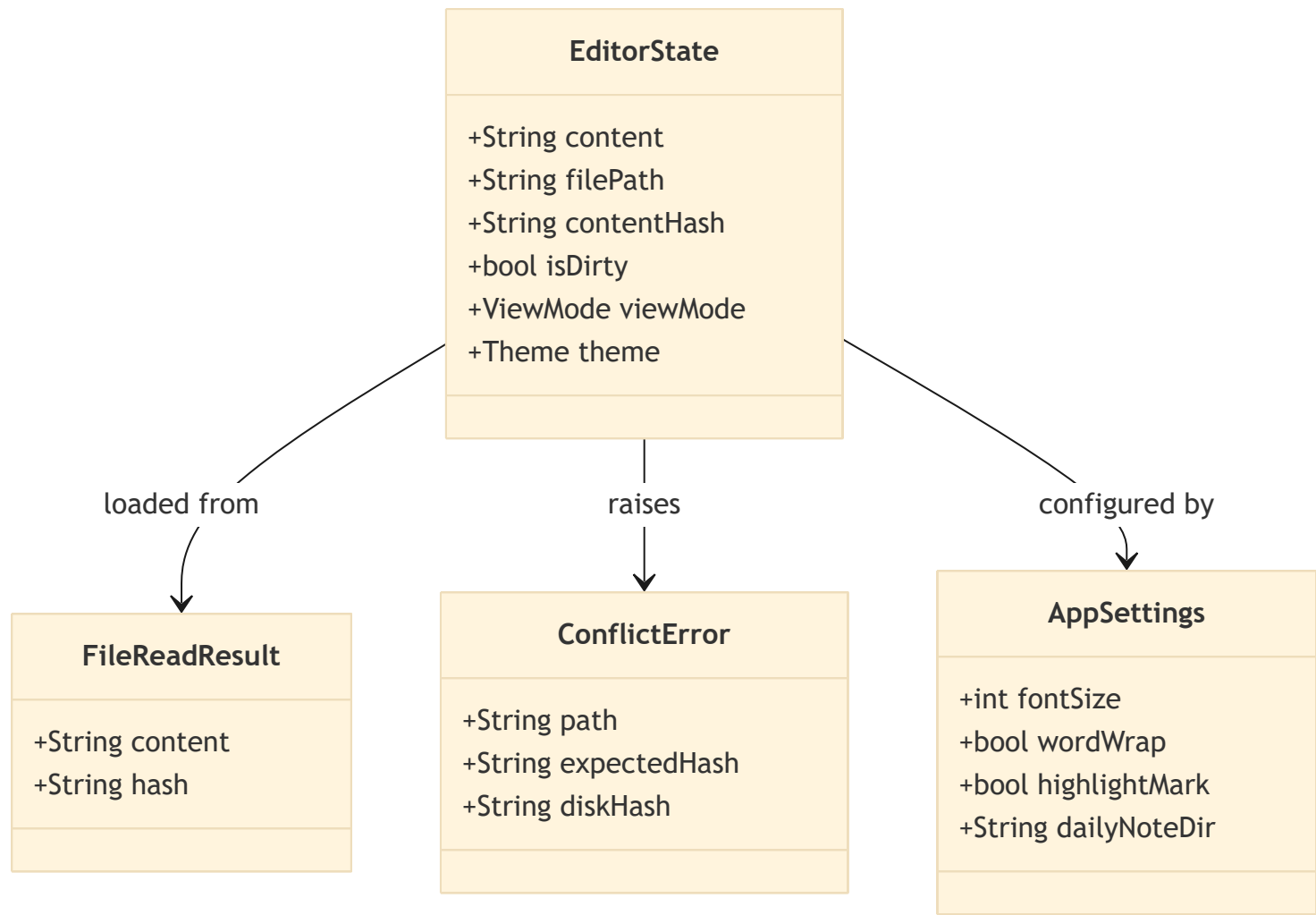
Flowchart — Lattice Edit Cycle



Sequence Diagram — File Open Flow

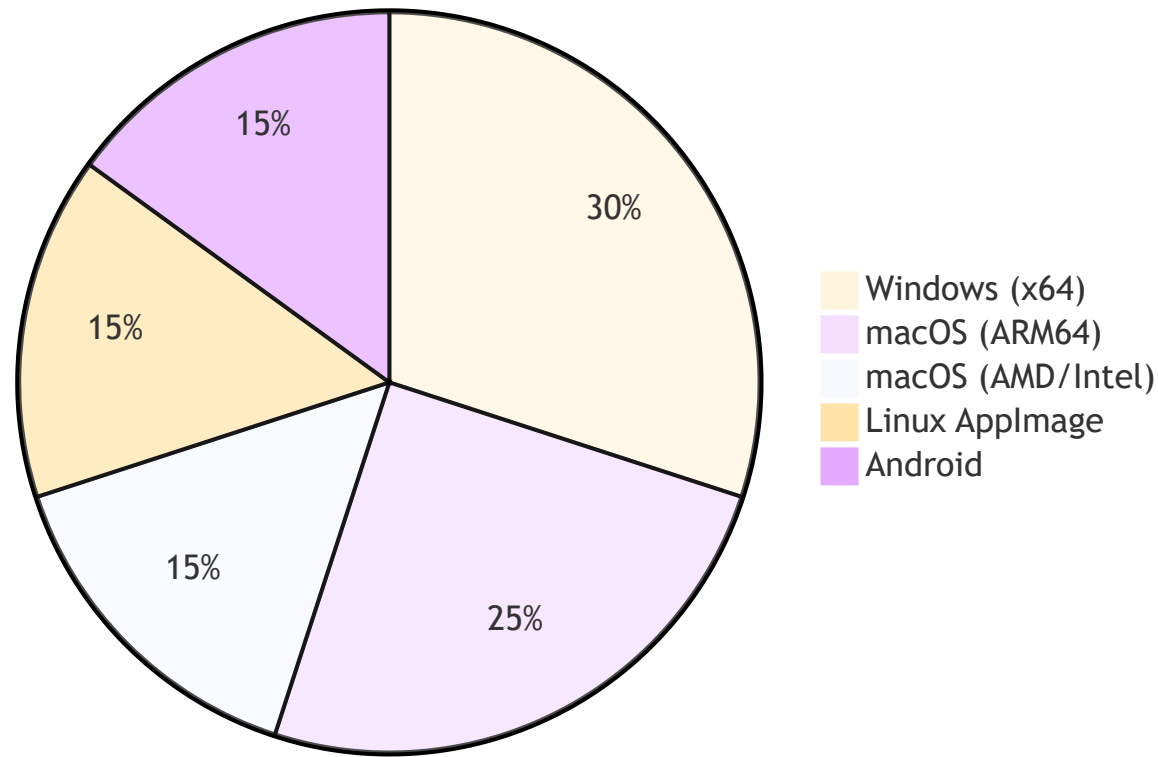


Class Diagram — Core Data Model



Pie Chart — Supported Platforms

Lattice Platform Coverage



Images

Paste images directly into Lattice with `Ctrl/Cmd+V` .
The image file is automatically saved to a sibling folder named after the current document, and a Markdown reference is inserted at the cursor position:

```
![[Image](demo_assets/img_1780574419628.png)]  
  
![[Image](demo_assets/img_1780575215039.png)]
```

Here is how it renders in Dual (edit on the left) mode:

demo.md - lattice (v0.2.39 · windows-desktop-2026W23.0EEC)

%USERPROFILE%_prj\lattice-ws\lattice\docs\demo\demo.md

207 v ## Mathematics — KaTeX

208

209 Lattice renders mathematics via **KaTeX**. Use ``$...$`` for inline and ``$$...$$`` for display math.

210

211 v ### Inline math

212

213 v Einstein's famous equation $E = mc^2$ and the quadratic formula

214 $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ render inline.

215

216 The Euler identity $e^{i\pi} + 1 = 0$ is often called the most beautiful equation in mathematics.

217

218 v ### Display math — Calculus

219

220 v
$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

221
$$\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

222
$$\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

223

224 v
$$\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

225
$$\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

226
$$\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

227

228 v ### Display math — Linear Algebra

229

230 v
$$\mathbf{A} \mathbf{x} = \lambda \mathbf{x} \implies \det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

231
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

232
$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

233
$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

234
$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

235

236 v
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

237
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

238
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

239
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

240
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

241

242 v ### Display math — Series & Limits

243

244 v
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

245
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

246
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

247
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

248

249 v
$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

250
$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

251
$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

252

253 ---

254

Mathematics — KaTeX

Lattice renders mathematics via **KaTeX**. Use ``$...$`` for inline and ``$$...$$`` for display math.

Inline math

Einstein's famous equation $E = mc^2$ and the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ render inline.

The Euler identity $e^{i\pi} + 1 = 0$ is often called the most beautiful equation in mathematics.

Display math — Calculus

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$
$$\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

Display math — Linear Algebra

$$\mathbf{A} \mathbf{x} = \lambda \mathbf{x} \implies \det(\mathbf{A} - \lambda \mathbf{I}) = 0$$
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Display math — Series & Limits

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$
$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

Mermaid Diagrams

Lattice renders ````mermaid```` fenced blocks inline in the preview.

Flowchart — Lattice Edit Cycle

Here is how it renders in Dual (edit on the bottom) mode:

Page 7 of 9

Lattice renders mathematics via **KaTeX**. Use $\dots$ for inline and
$$\dots$$
 for display math.

Einstein's famous equation $E = mc^2$ and the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ render inline.

The Euler identity $e^{i\pi} + 1 = 0$ is often called the most beautiful equation in mathematics.

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$
$$\mathbf{Ax} = \lambda \mathbf{x} \implies \det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

```

205 ---
206
207 ## Mathematics - KaTeX
208
209 Lattice renders mathematics via **KaTeX**. Use `$...$` for inline and `$$...$$` for display math.
210
211 ### Inline math
212
213 Einstein's famous equation  $E = mc^2$  and the quadratic formula
214  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$  render inline.
215
216 The Euler identity  $e^{i\pi} + 1 = 0$  is often called the most beautiful equation in mathematics.
217
218 ### Display math - Calculus
219
220 
$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

221
222 
$$\frac{d}{dx} \left[ \int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

223
224 
$$\mathbf{A} \mathbf{x} = \lambda \mathbf{x}$$

225 
$$\quad \Longleftrightarrow$$

226 
$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

227
228 ### Display math - Linear Algebra
229
230 
$$\mathbf{A} \mathbf{x} = \lambda \mathbf{x}$$

231 
$$\quad \Longleftrightarrow$$

232 
$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

233

```

Page 8 of 9

Links

- [Lattice on GitHub](#) — source code and issue tracker
- [Releases Page](#) — download installers
- [Release Site](#) — GitHub Pages release portal
- Relative link: [System Architecture Demo](#)
- Relative link: [Physics Equations Demo](#)

Auto-link: <https://github.com/blessia-blessini/lattice>

HTML in Markdown

Lattice renders a **safe whitelist** of inline HTML tags and passes everything else through as literal text. The linter flags all inline HTML — whitelisted tags get a **hint** (portability risk), non-whitelisted tags get a **warning** (not rendered here).

HTML element	Lattice preview	GitHub	Portable alternative
<sup>	✓ rendered	✓	KaTeX: x^2 → X^2 · Unicode: x²
<sub>	✓ rendered	✓	KaTeX: H_{20} → H_2O · Unicode: H₂O
<kbd>	✓ rendered	✓	Inline code: <code>Ctrl+S</code>
 	✓ rendered	✓	Two trailing spaces or blank line
<mark>	⚠ plain text	✓	<code>==highlight==</code> (Lattice-native)
<details>	⚠ plain text	✓	Headings + TOC (no universal equivalent)
Other HTML	⚠ plain text	varies	Markdown equivalent (case-by-case)

Live examples of the whitelisted tags:

- Superscript: $X^2 + Y^3$
- Subscript: H₂O and CO₂
- Keyboard: press `Ctrl` + `S` to save

Keyboard Shortcuts Reference

Action	Windows / Linux	macOS
Save	<code>Ctrl+S</code>	<code>Cmd+S</code>
Open file	<code>Ctrl+O</code>	<code>Cmd+O</code>
New window	<code>Ctrl+N</code>	<code>Cmd+N</code>
Generate / refresh TOC	<code>Ctrl+Shift+T</code>	<code>Cmd+Shift+T</code>
Auto-align table	<code>Ctrl+Shift+L</code>	<code>Cmd+Shift+L</code>
Paste image from clipboard	<code>Ctrl+V</code>	<code>Cmd+V</code>
Print / Save as PDF	<code>Ctrl+P</code>	<code>Cmd+P</code>
Increase font size	<code>Ctrl++</code>	<code>Cmd++</code>
Decrease font size	<code>Ctrl+-</code>	<code>Cmd+-</code>